of the 2-sphere and the torus is a 2-cycle, and there are no 2-boundaries, so $H_2(X) \simeq \mathbb{Z} \times \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

11. The space X is connected, so $H_0(X) \simeq \mathbb{Z}$. Each of the two handles has a 1-cycle that is a circle around the handle, like the circle a in Fig. 42.13. We let a be such a 1-cycle on the left handle and a' an analogous one on the right handle. Then we have 1-cycles b on the left handle and b' on the right handle, which go along the handle, onto the sphere, and then back on the handle at its other end. These are analogous to the circle b on the annulus in Fig. 42.13. We should also consider the 1-cycle z that is the "equator" of the sphere in Fig. 42.18, going through the holes made by the handles. If we cut the space along this equatorial circle z, it does not fall into two pieces; the handles hold it together, and the boundary of the resulting 2-figure consists of two copies of the circle with opposite orientation, whose algebraic sum in $C_n(X)$ is then zero, so the equatorial circle z is not a 1-boundary. However, if we slice the entire figure into two pieces with a horizontal slash which cuts the sphere on the equatorial circle z and cuts the handles at their extreme left and right points, the space does fall into two pieces, and the boundary of each consists the equatorial circle z and two 1-cycles of type a and a'. Thus, assuming proper orientation, we have $(z + a + a') \in B_1(X)$, so z is in the coset $(-a - a') + B_1(X)$. Thus the cosets $(ma + m'a' + nb + n'b') + B_1(X)$ include the 1-cycles homologous to z, and we see that $H_1(X) \simeq \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$. The 2-cycles consist of sums of multiples of the entire space, so $H_2(X) \simeq \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

12. The space X is connected, so $H_0(X) \simeq \mathbb{Z}$. An analysis similar to the one we made in the solution of Exercise 42.11 indicates that $H_1(X) \simeq \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{2n \text{ factors}}$. Multiples of the entire space are the only 2-cycles so $H_2((X) \simeq \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

43. More Homology Computations and Applications

1. *2-triangle triangulation:* $n_0 = 4, n_1 = 5, n_2 = 2, n_3 = 0$.
$\chi(X) = 4 - 5 + 2 - 0 = 1$.

6-triangle triangulation: $n_0 = 7, n_1 = 12, n_2 = 6, n_3 = 0$.
$\chi(X) = 7 - 12 + 6 - 0 = 1$.

2. **a.** From Fig. 42.11 we have $n_0 = 10, n_1 = 20, n_2 = 10$, and $n_3 = 0$ so $\chi(X) = 10 - 20 + 10 - 0 = 0$. From the homology groups computed in Example 42.10, we have $\beta_0 = 1, \beta_1 = 1, \beta_2 = 0$, and $\beta_3 = 0$ so $\beta_0 - \beta_1 + \beta_2 - \beta_3 = 1 - 1 + 0 - 0 = 0$.

b. From Fig. 42.14 we have $n_0 = 9, n_1 = 27, n_2 = 18$, and $n_3 = 0$ so $\chi(X) = 9 - 27 + 18 - 0 = 0$. From the homology groups computed in Example 42.13, we have $\beta_0 = 1, \beta_1 = 2, \beta_2 = 1$, and $\beta_3 = 0$ so $\beta_0 - \beta_1 + \beta_2 - \beta_3 = 1 - 2 + 1 - 0 = 0$.

c. Taking the triangulation in Fig. 42.14 with the arrow on the top edge reversed for the Klein bottle, we see by Part(**b**) that $\chi(X) = 0$. From the homology groups computed in Example 43.1, we have $\beta_0 = 1, \beta_1 = 1, \beta_2 = 0$, and $\beta_3 = 0$, so $\beta_0 - \beta_1 + \beta_2 - \beta_3 = 1 - 1 + 0 - 0 = 0$.

3. The theorem will hold for a square region, for such a region is homeomorphic to E^2. It obviously does not hold for two disjoint 2-cells, for each can be mapped continuously into the other, and such a map has no fixed points.

4. The space X is connected so $H_0(X) \simeq \mathbb{Z}$. The 2-sphere contains no 1-cycles that are not 1-boundaries, while the Klein bottle has 1-cycles mb for all $m \in \mathbb{Z}$, none of which are 1-boundaries, and a 1-cycle a

which is not a 1-boundary although $2a$ is a 1-boundary, as shown in Example 43.1 Thus we see that $H_1(X) \simeq \mathbb{Z}_2 \times \mathbb{Z}$. The 2-sphere is a 2-cycle, while the Klein bottle is not, as shown in Example 43.1. Thus $H_2(X) \simeq \mathbb{Z}$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

5. If P_1 is a vertex on one Klein bottle and P_2 is a vertex on the other, then there is no 1-cycle with boundary $P_2 - P_1$, so the 0-cycles are the elements of the cosets $(mP_1 + nP_2) + B_0(X)$ for $m, n \in \mathbb{Z}$. Thus $H_0(X) \simeq \mathbb{Z} \times \mathbb{Z}$. Each Klein bottle contributes a homology class $b + B_1(X)$ of infinite order in $H_1(X)$ and a homology class $a + B_1(X)$ of order 2, as explained in Example 43.1. Thus $H_1(X) \simeq \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z} \times \mathbb{Z}$. Neither Klein bottle is a 2-cycle, as explained in Example 43.1, so $H_2(X) = 0$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

6. F T T F T T T T T F

7. By Exercise 12 of Section 42, the nonzero homology groups of a 2-sphere with n handles are $H_0(X) \simeq \mathbb{Z}, H_1(X) \simeq \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{2n \text{ factors}}$, and $H_2(X) \simeq \mathbb{Z}$. Using Theorem 43.7, we obtain $\beta_0 = 1, \beta_1 = 2n$, and $\beta_2 = 1$, so $\chi(X) = \beta_0 - \beta_1 + \beta_2 = 1 - 2n + 1 = -2n + 2$.

8. We describe a trianglation. Viewing the circle in Fig. 43.14 as the usual face of a 12 hour clock, mark points Q on the circle at 10:30 and 4:30, and mark points R at 1:30 and 7:30. Draw lines joining the two points P, joining the two points Q, and joining the two points R. They meet at a vertex C in the center of the circle. Then draw the lines from Q at 10:30 to P at 3:00, from P at 3:00 to R at 7:30, and from R at 7:30 to Q at 10:30; they contribute 3 more vertices where lines intersect. This gives us a triangulation of the projective plane.

Because the projective plane X is connected, $H_0(X) \simeq \mathbb{Z}$. Every 1-cycle can be "pushed" (by subtracting 1-boundaries) to the arcs on the circle. More specifically, starting with the triangle having as bottom edge the line from C to 3:00 P and going around counterclockwise, we can eliminate that bottom edge, the right edge of the next triangle having C as vertex, etc., until we have eliminated the edge from C to 7:30 R. This leaves the line from C to 4:30 Q. Then we start with some triangle having an arc of the circle as edge and go around counterclockwise in a similar way, eliminating the righthand edges. We are left with our 1-cycle having edges only on the arcs except for one or two single edges sticking in from the circle or out from the center, which also must have coefficient 0 in our 1-cycle, so really all edges are on the circle; note there are only three of them, PR, RQ and QP, not six of them. Because it is a 1-cycle, they must all occur with the same coefficients. Let $a = PR + RQ + QP$, as indicated on your figure. Now the boundary of the projective plane X is clearly $2a$, due to the same counterclockwise orientation of the two arcs labeled a. Thus $2a$ is a 1-boundary, and we see that $H_1(X) \simeq \mathbb{Z}_2$. Let c be the 2-chain consisting of the sum of all 2-simplexes oriented the same way. Because the boundary of $\partial_2(c) = 2a$, we see that $Z_2(X) = 0$ so $H_2(X) = 0$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

9. The space X is connected so $H_0(X) \simeq \mathbb{Z}$. Computation of $H_1(X)$ is our toughest job so far. As suggested in the exercise, view the space as the disk in Fig. 21.8 with $q - 1$ circular holes in it to be sewn up by sewing diametrically opposite points. Let us number these holes from 1 to $q - 1$, and let a_i be the top semicircle of the ith hole with counterclockwise orientation. Note that in view of the identification to take place in the sewing, we can also consider the bottom semicircle of the ith hole to be a_i with counterclockwise orientation. We consider the top and bottom semicircles of the rim of the disk in Fig. 21.8 to be a_q rather than a. They will form the qth crosscap in the final sewing. Now consider a circle on the 2-sphere that encircles only the ith crosscap of X for $i < q$. This corresponds to a circle going around the ith hole but not containing any other hole in our disk model of X. Without

making a triangulation, we can consider this circle, oriented counterclockwise, to be a 1-cycle z_i. Now z_i is not a 1-boundary, for if we cut our sphere along this circle, it falls into two pieces, neither of which has boundary z_i on account of the crosscaps contained in each piece. The piece that contains the ith crosscap, corresponding to the piece containing the ith hole in our disk model, has boundary $z_i - 2a_i$. Doing this for each i where $1 \leq i \leq q-1$, we obtain cycles z_i such that $(z_i - 2a_i) \in B_1(X)$ for $i = 1, 2, \cdots, q-1$. (Remember that z_i could be *any* 1-cycle encircling only crosscap number i on the 2-sphere X.) Let c be the 2-chain consisting of the sum of all 2-simplexes oriented the same way. Another element of $B_1(X)$ is $\partial_2(c)$, and Exercise 8 indicates that $\partial_2(c) = 2a_1 + 2a_2 + \cdots + 2a_q$. As indicated in Exercise 8, each a_i is a cycle for $i = 1, 2, \cdots, q$, but no a_i is a 1-boundary, and indeed no ma_i is a 1-boundary. It is only the *sum* of all the $m(2a_i)$ that is a 1-boundary for all $m \in \mathbb{Z}$.

We take as generators for the 1-cycles in X the 1-cycles $a_1, a_2, \cdots, a_{q-1}$, and $a_1 + a_2 + \cdots + a_q$. Note that the coset $2ma_i + B_1(X)$ contains the 1-cycle mz_i for all $m \in \mathbb{Z}$, because $mz_i - 2ma_i = m(z_i - 2a_i) \in B_1(X)$. Since the generators a_i have infinite order for $1 \leq i \leq q-1$ and the generator $a_1 + a_2 + \cdots + a_q$ has order 2, we see that $H_1(X) \simeq \underbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}_{q\text{-1 factors}} \times \mathbb{Z}_2$.

Let c be the 2-chain consisting of the sum of all 2-simplexes oriented the same way. Because $\partial_2(c) \neq 0$, there are no 2-cycles so $H_2(X) = 0$. There is nothing of higher dimension, so $H_i(X) = 0$ for $i > 2$.

You may be bothered by the apparent lack of symmetry in our computation of $H_1(X)$. What was special about the qth crosscap, that didn't have a 1-cycle z_q around it? The answer is, it did. Remember that z_i could be any 1-cycle containing the ith crosscap. We can take for z_1 and z_2 two 1-cycles that have an edge in common. In their sum, $z_1 + z_2$, that edge cancels and we get a 1-cycle containing both the first and second crosscaps. Continuing in the obvious way, we can get a cycle $z_1 + z_2 + \cdots + z_{q-1}$ containing all the crosscaps but the qth. Viewed on the 2-sphere, we realize that we can view this as a 1-cycle $-z_q$ that contains only the qth crosscap. Note that from the 1-boundaries $z_i - 2a_i$ for $i = 1, 2, \cdots, q-1$ and $2a_1 + 2a_2 + \cdots + 2a_q$ that we found in $B_1(X)$, we can deduce that $B_1(X)$ contains $(z_1 - 2a_1) + (z_2 - 2a_2) + \cdots + (z_{q-1} - 2a_{q-1}) + (2a_1 + 2a_2 + \cdots + 2a_q) = -z_q + 2a_q$, so $z_q - 2a_q$ is in $B_1(X)$ and everything really is symmetric.

10. Let Q be a vertex of X on a, and let c be the 2-chain consisting of all 2-simplexes of X, all oriented the same way, so that $c \in Z_2(X)$.

a. $f_{*0} : H_0(X) \to H_0(X)$ is given by

$$f_{*0}(Q + B_0(X)) = Q + B_0(X),$$

that is, f_{*0} is the identity map.

$f_{*1} : H_1(X) \to H_1(X)$ is given by

$$f_{*1}((ma + nb) + B_1(X)) = (ma + 2nb) + B_1(X),$$

reflecting the fact that f maps X twice around itself in the θ direction.

$f_{*2} : H_2(X) \to H_2(X)$ is given by

$$f_{*2}(c + B_2(X)) = 2c + B_2(X),$$

reflecting the fact that each point of X is the image of two points of X under f.

b. f_{*0} is as in Part(**a**).

$f_{*1} : H_1(X) \to H_1(X)$ is given by

$$f_{*1}((ma + nb) + B_1(X)) = (2ma + nb) + B_1(X),$$

reflecting the fact that f maps X twice around itself in the ϕ direction.

f_{*2} is as in Part(**a**).

c. f_{*0} is as in Part(**a**).

$f_{*1} : H_1(X) \to H_1(X)$ is given by

$$f_{*1}((ma + nb) + B_1(X)) = (2ma + 2nb) + B_1(X),$$

reflecting the fact that f maps X twice around itself in both the θ and the ϕ directions.

$f_{*2} : H_2(X) \to H_2(X)$ is given by

$$f_{*2}(c + B_2(X)) = 4c + B_2(X),$$

reflecting the fact that each point of X is the image of four points of X under f.

11. Let Q be a vertex of b, and let c be the 2-chain consisting of all 2-simplexes of X, all oriented the same way, so that $c \in Z_2(X)$.

a. $f_{*0} : H_0(X) \to H_0(b)$ is given by $f_{*0}(Q + B_0(X)) = Q + B_0(b)$.

$f_{*1} : H_1(X) \to H_1(b)$ is given by $f_{*1}((ma + nb) + B_1(X)) = nb + B_1(b)$.

$f_{*2} : H_2(X) \to H_2(X)$ is given by $f_{*2}(c + B_2(X)) = 0$.

b. f_{*0} is as in Part(**a**).

$f_{*1} : H_1(X) \to H_1(b)$ is given by $f_{*1}((ma + nb) + B_1(X)) = 2nb + B_1(b)$.

f_{*2} is as in Part(**a**).

12. The answers are the same as for Exercise 11 with $B_i(b)$ replaced by $B_i(X)$.

13. Let Q be a vertex on b.

$f_{*0} : H_0(X) \to H_0(b)$ is given by $f_{*0}(Q + B_0(X)) = Q + B_0(b)$.

$f_{*1} : H_1(X) \to H_1(b)$ is given by $f_{*1}((ma + nb) + B_1(X)) = nb + B_1(b)$, where $m = 0, 1$.

f_{*2} is trivial, because both $H_2(X)$ and $H_2(b)$ are 0.